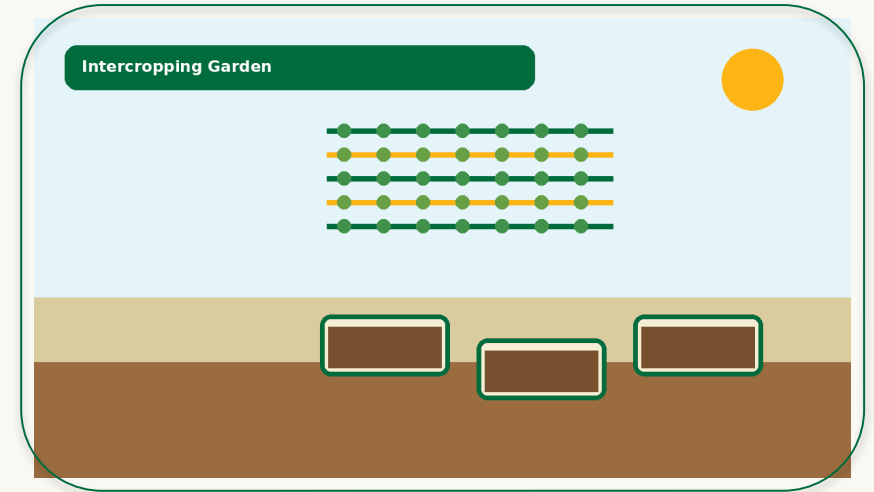


Intercropping Garden

Crop Interactions and Land Use

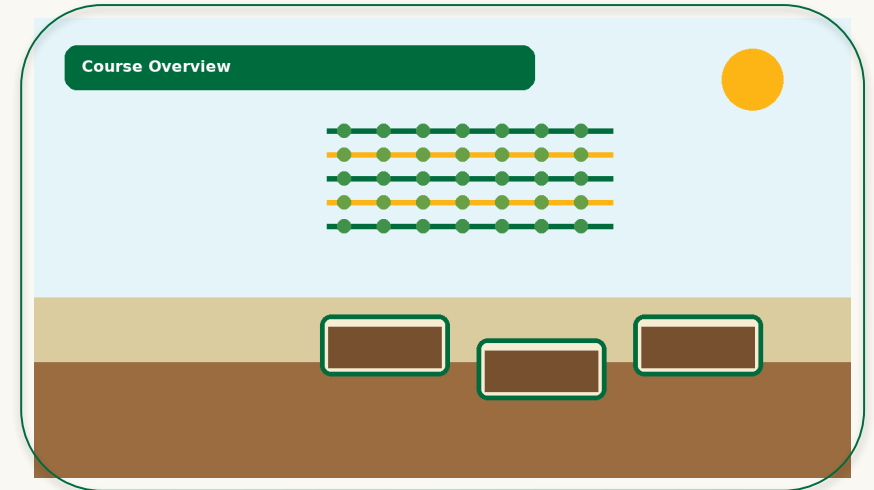
- LIFEWS AgriShine™ GrowMeal™ JSS 1-3 Teacher Training
- Audience: JSS 1-3 teachers, agricultural-science educators and garden facilitators
- Format: Self-paced online course + supervised practical/laboratory orientation
- Recommended duration: 3.5 hours online + 2 hours practical
- JSS learning progression: formulate -> test -> measure -> analyze -> manage



Course Overview

Learning outcomes, materials and teacher readiness

- Learning outcomes: explain the production or experimental system; teach all ten technical topics accurately; guide students in planning, measurement and data collection; connect agricultural science with mathematics and evidence; supervise field activities safely; maintain credible records; help students distinguish observation, analysis and conclusion.
- Core materials: teacher guide, plot/system map, crop plan, child-safe field tools, irrigation equipment, labels, clipboards, measuring tapes/rulers, scales and sensors where relevant, data sheets, clean harvest containers, PPE/handwashing supplies and first-aid access.
- Completion expectation: complete all modules, demonstrate one safe practical session, and submit a lesson, production plan or experimental protocol appropriate to the course.



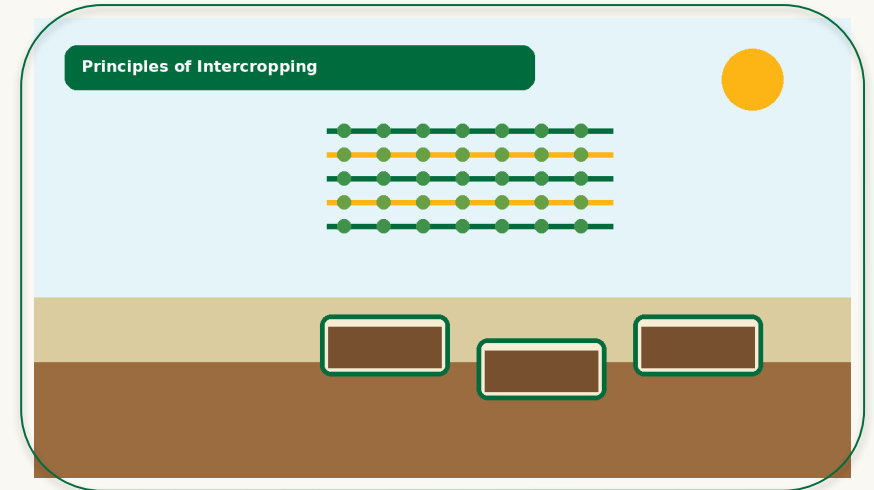
TEACHER PRACTICE

- Training sequence: QUESTION -> PLAN -> IMPLEMENT -> MEASURE -> ANALYZE -> DECIDE.

Principles of Intercropping

Teacher knowledge, technical practice and student supervision

- Define intercropping as growing two or more crops in the same field area during overlapping periods.
- Explain that system design aims to use resources, space or time in complementary ways while managing competition.
- Distinguish intentional intercropping from random mixed planting.
- Identify the component crops and arrangement clearly.
- Use locally relevant crop combinations supported by agronomic guidance.



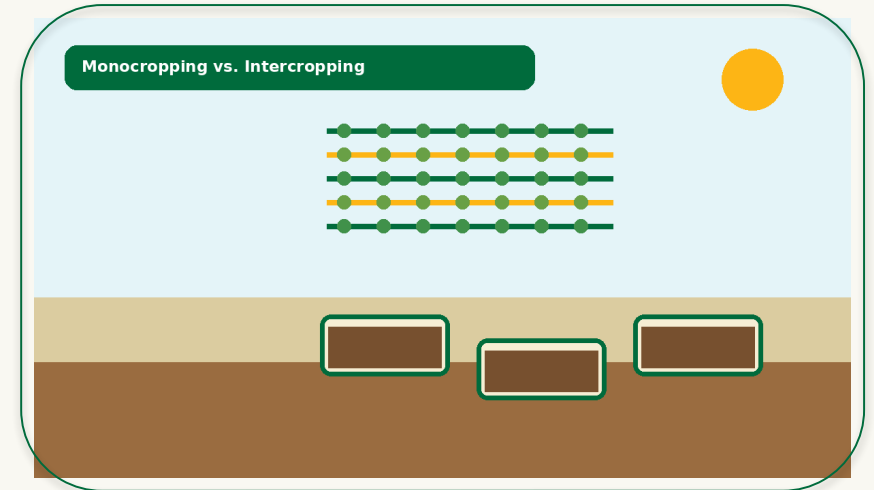
TEACHER PRACTICE

- Describe one planned intercrop and its management objective.

Monocropping vs. Intercropping

Teacher knowledge, technical practice and student supervision

- Define monocropping for the experiment as one crop grown alone in the comparison plot.
- Use monocrop plots of each component crop when evaluating intercrop performance where space permits.
- Keep planting date, management and measurement methods comparable.
- Discuss practical differences in weeding, harvest and crop protection.
- Avoid assuming intercropping automatically yields more.



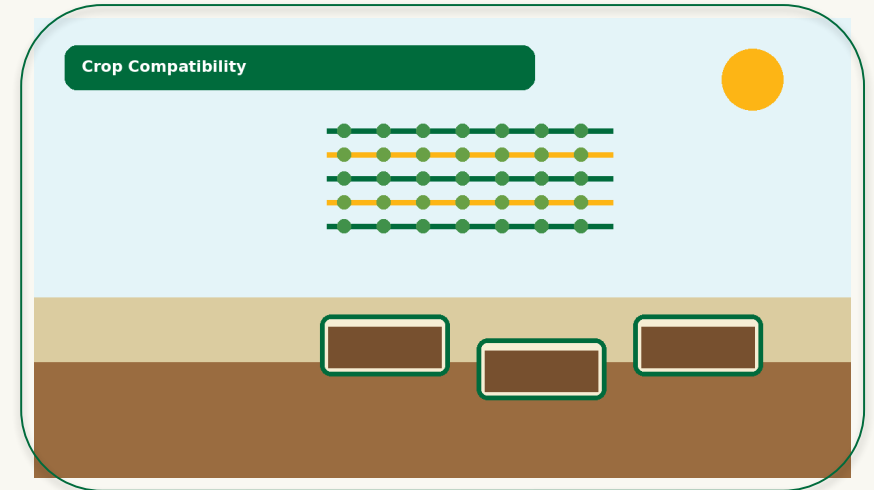
TEACHER PRACTICE

- Design a fair monocrop-intercrop comparison.

Crop Compatibility

Teacher knowledge, technical practice and student supervision

- Consider crop height, rooting pattern, duration, nutrient demand, canopy structure and management requirements.
- Avoid combinations with severe shading or incompatible harvest operations.
- Use local recommendations and previous school records where available.
- Recognize compatibility can change with spacing and season.
- State the reason for selecting each crop pair.



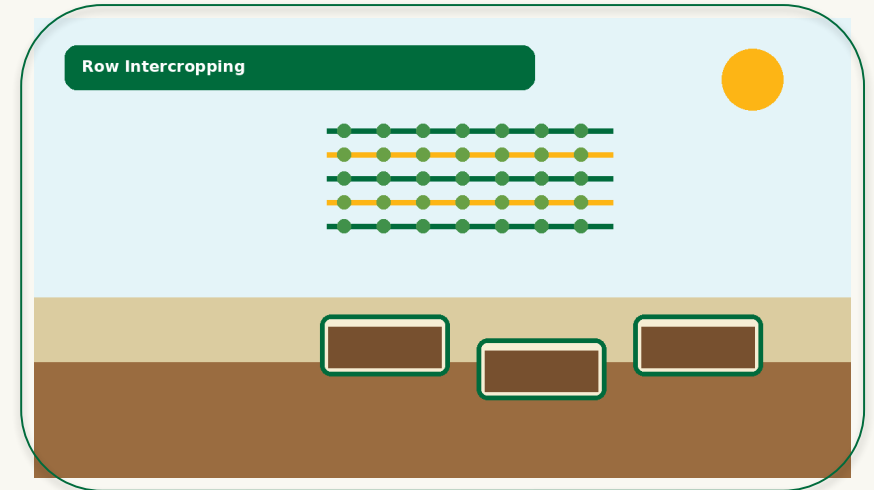
TEACHER PRACTICE

- Prepare a crop-compatibility matrix.

Row Intercropping

Teacher knowledge, technical practice and student supervision

- Arrange component crops in distinct alternating rows.
- Define row ratio, within-row spacing and total plot dimensions.
- Maintain crop-specific populations as intended by the experimental design.
- Use row labels to support measurement.
- Observe edge and shading effects.



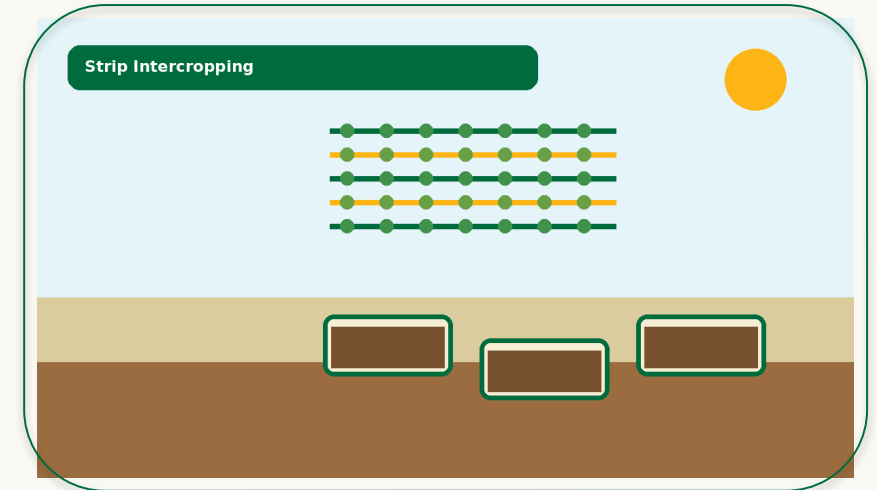
TEACHER PRACTICE

- Lay out a 1:1 or 2:1 row-intercrop plan.

Strip Intercropping

Teacher knowledge, technical practice and student supervision

- Define strips as wider bands of component crops managed within the same field area.
- Choose strip widths that remain manageable at school scale.
- Maintain clear boundaries for data collection.
- Compare strip arrangement with row intercropping conceptually or experimentally.
- Consider access and irrigation across strips.



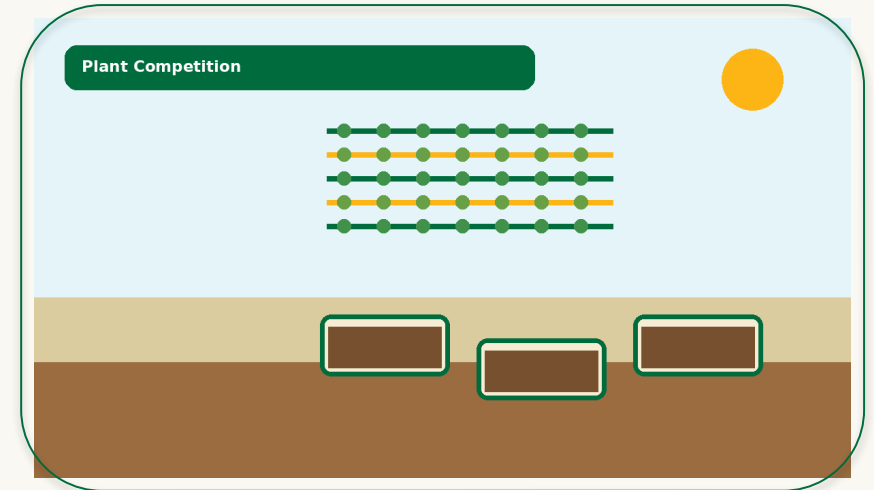
TEACHER PRACTICE

- Sketch a safe school-scale strip-intercrop layout.

Plant Competition

Teacher knowledge, technical practice and student supervision

- Explain competition for light, water, nutrients and space when crop demands overlap.
- Observe canopy shading, soil moisture or growth differences as possible indicators.
- Avoid claiming competition from one visual symptom alone.
- Use measurements to test suspected effects.
- Adjust spacing only according to the planned design.



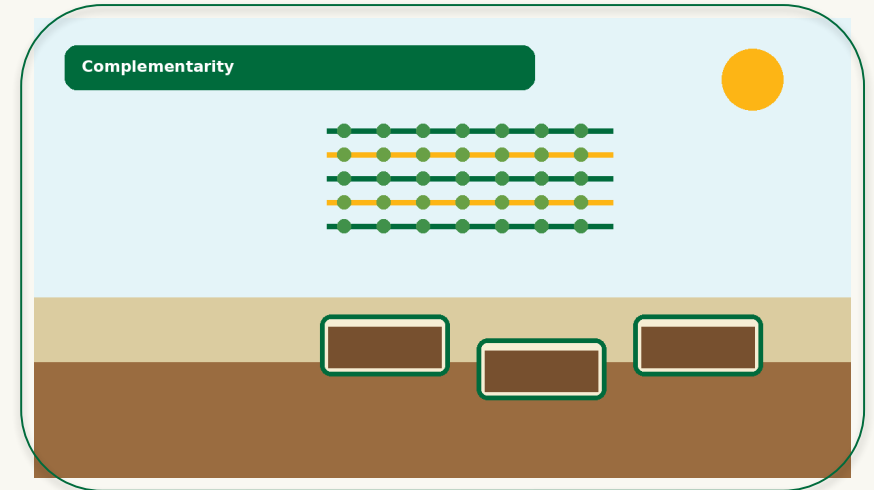
TEACHER PRACTICE

- Identify two measurable indicators of competition.

Complementarity

Teacher knowledge, technical practice and student supervision

- Explain complementarity as component crops using resources differently in ways that can reduce direct competition or improve combined system performance.
- Use examples such as different canopy heights or growth durations without guaranteeing a benefit.
- Distinguish complementarity from facilitation where advanced discussion is appropriate.
- Measure rather than assume complementary effects.
- Relate observations to the original design rationale.



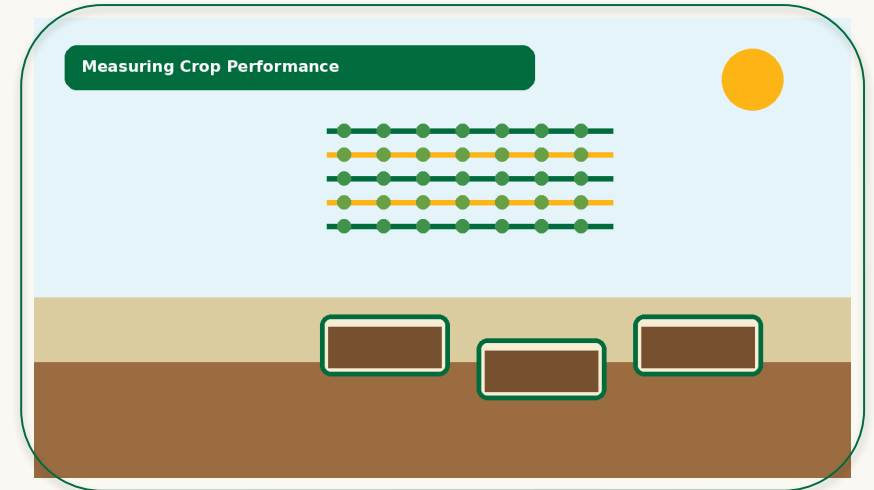
TEACHER PRACTICE

- Write a testable complementarity prediction.

Measuring Crop Performance

Teacher knowledge, technical practice and student supervision

- Measure each component crop separately for survival, growth and harvest.
- Keep sample rules consistent across monocrop and intercrop plots.
- Record plant population because population differences affect yield comparisons.
- Measure relevant resource indicators where feasible.
- Document pest, lodging or shading events.



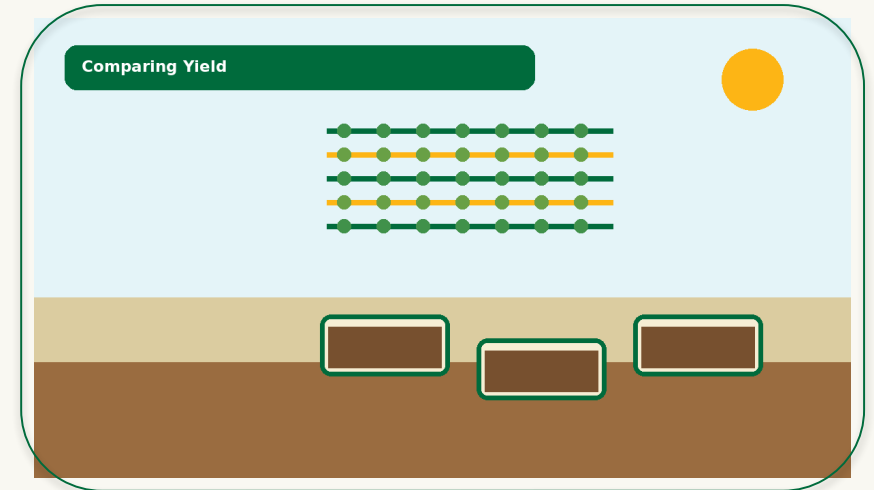
TEACHER PRACTICE

- Create a component-specific performance sheet.

Comparing Yield

Teacher knowledge, technical practice and student supervision

- Record yield of each component crop in consistent units and known plot areas.
- Compare component yields with their respective monocrops.
- Calculate total harvested mass cautiously because crops differ in value and biology.
- Introduce land-equivalent concepts only if appropriate and clearly taught.
- Do not interpret one season as universal evidence.



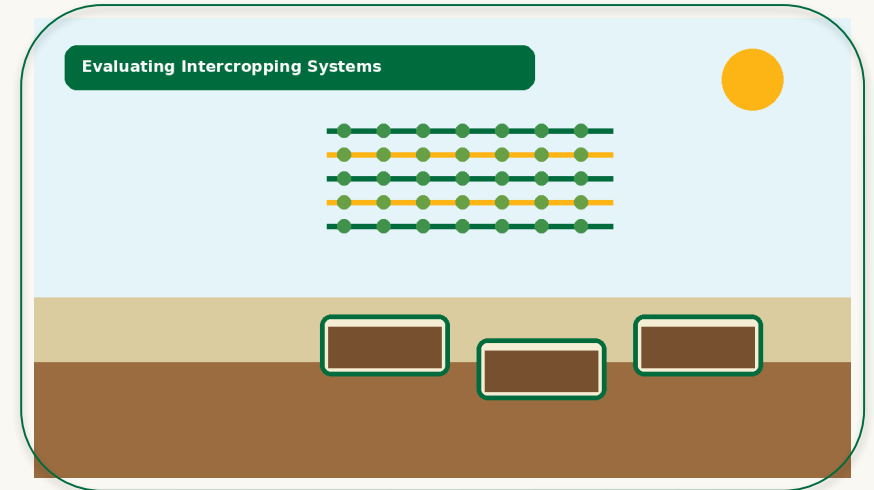
TEACHER PRACTICE

- Prepare a yield-comparison table for two crops.

Evaluating Intercropping Systems

Teacher knowledge, technical practice and student supervision

- Evaluate combined yield, component performance, land use, water/management demand, labor and harvest practicality.
- Discuss benefits and trade-offs rather than seeking a single 'winner'.
- Use experimental evidence and production records.
- Identify design changes that could improve the next cycle.
- State conclusions within the limits of the school trial.



TEACHER PRACTICE

- Completion task: produce an evidence-based intercrop system evaluation.